

Data Analysis Project I Report

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1. Introduction

This report analyzes employee salary data to understand salary differences between genders and employment categories.

The dataset contains 474 employees and includes the following variables:

- Gender
- Employment Category
- Current Salary

The objective is to explore salary patterns and relationships between these variables.

2. Dataset Description:

The dataset consists of 474 observations.

Variables include:

Gender -> categorical variable

Employment Category -> categorical variable

Current Salary -> numerical variable

Gender distribution:

- 216 females
- 258 males

3. Salary Analysis by Gender:

From the descriptive statistics:

Female salary statistics

Mean salary: 26,031.92

Median: 24,300

Standard deviation: 7,558

Minimum: 15,750

Maximum: 58,125

Male salary statistics

Mean salary: 41,441.78

Median: 32,850

Standard deviation: 19,499

Minimum: 19,650

Maximum: 135,000

Descriptive statistics show significant salary differences between male and female employees.

The average salary for females is \$26,031, while the average salary for males is \$41,442.

This indicates that males earn approximately \$15,000 more on average.

Male salaries also show much higher variability with a standard deviation of \$19,499, compared to \$7,558 for females.

The maximum salary for males reaches \$135,000, while the maximum female salary is \$58,125.

This suggests that high-paying positions are more commonly held by males.

4. Relationship Between Gender and Employment Category:

From the crosstab results:

Total employees: 474

Gender distribution:

Female: 216 divided into:

- 95.4% work in clerical positions
- 0% work in custodial positions
- 4.6% work in managerial positions

Male: 258 divided into:

- 60.9% work in clerical positions
- 10.5% work in custodial positions
- 28.7% work in managerial positions

Percentages:

Female employees: 45.6%

Male employees: 54.4%

The dataset shows that 54.4% of employees are male and 45.6% are female.

Female employees are mainly concentrated in clerical positions, while male employees are more evenly distributed across all employment categories.

Overall, 76.6% of employees work in clerical roles, 17.7% in managerial roles, and 5.7% in custodial roles.

This shows that clerical jobs dominate the workforce, followed by managerial roles, with custodial positions being the least common.

In addition, men are more represented in managerial positions, while women are mostly found in clerical jobs.

5. Salary by Employment Category:

Employment categories in the dataset:

- Clerical
- Custodial
- Managerial

These represent different job levels.

Typical salary pattern:

Managers -> highest salary

Clerical -> moderate salary

Custodial -> lower salary

Clerical employees:

The average salary is \$27,838.54, with a median of \$26,550. Salaries range from \$15,750 to \$80,000, and the standard deviation is \$7,567.995, indicating moderate variation in pay.

Custodial employees:

The average salary is \$30,938.89, with a median of \$30,750. Salaries range from \$24,300 to \$35,250, and the standard deviation is about \$2,114, showing relatively low variation in salaries.

Managerial employees:

The average salary is \$63,977.80, with a median of \$60,500. Salaries range from \$34,410 to \$135,000, and the standard deviation is \$18,244, indicating a wide variation in managerial salaries.

6. Conclusion:

The analysis shows a clear difference in salaries across both gender and employment category.

Male employees earn more on average than female employees, and they are also more likely to be in managerial positions.

Female employees are mostly concentrated in clerical jobs, while male employees are distributed across clerical, custodial, and managerial roles.

Managerial positions have the highest salaries and the widest salary range, while custodial salaries are the most consistent.

Overall, the results suggest that employment category plays an important role in explaining the salary gap in this dataset.